

Netflix Data Analysis Using POWERBI

Project Title: Netflix Data Analysis Using POWERBI

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Domain: Entertainment, Media, and OTT Analytics

Functions: DAX (Data Analysis Expressions)

Tools: Excel, Microsoft Power BI

Problem Statement:

Analyze Netflix's content and customer viewing behavior to derive meaningful business insights. The goal is to clean and visualize the data to understand customer engagement, content popularity, and subscription trends.

Business Use Cases:

- Identify the most popular movies and TV shows on Netflix based on watch time and ratings.
- Analyze customer demographics to understand target audience preferences.
- Understand the impact of subscription types on watch time and customer engagement.
- Explore content ratings and their effect on viewer behavior.
- Provide insights on the best time for content release to maximize engagement.
- Assist Netflix in refining its content strategy based on genre popularity and user ratings.
- Compare customer retention and engagement patterns across different countries.

DataSet:

1. Netflix Shows Dataset - **Format:** CSV - **Key Variables:** show_id, type, title, director, cast, country, date_added, release_year, rating, duration, genre.

2. Netflix Customer Ratings Dataset - **Format:** CSV **Key Variables:** customer_id, show_id, customer_name, age, gender, country, subscription_type, watch_time_minutes, rating.

Data Set Explanation:

- **show_id (String):** Unique identifier for each movie or TV show.
- **type (String):** Specifies whether the content is a "Movie" or "TV Show."
- **title (String):** Name of the movie or TV show.
- **director (String):** Name of the director of the content (if available).
- **cast (String):** List of main actors in the content.
- **country (String):** The country where the content was produced.
- **date_added (Date):** Date when the content was added to Netflix.
- **release_year (Integer):** The year when the content was originally released.
- **rating (String):** Content rating such as "PG", "TV-MA", etc

Tools Used:

Excel:

Microsoft Excel is used for data entry, calculations, data cleaning, and basic analysis using tables, formulas, and charts.

Power BI:

Microsoft Power BI is used to create interactive dashboards, visualizations, and business reports for better decision-making.

Approach:

1.Data Cleaning & Transformation (Excel + Power BI)

- **Identify and fix missing values in director, cast, and country columns in the Netflix dataset.**
 - ✓ Checked for null or blank values using filters and data profiling.
 - ✓ Replaced missing values in:
 - **director** column with "Unknown Director"
 - **cast** column with "Not Available"
 - **country** column with "Unknown Country"
- **Standardize date formats in the date_added column.**
 - ✓ Converted the **date_added** column into a standard date format.

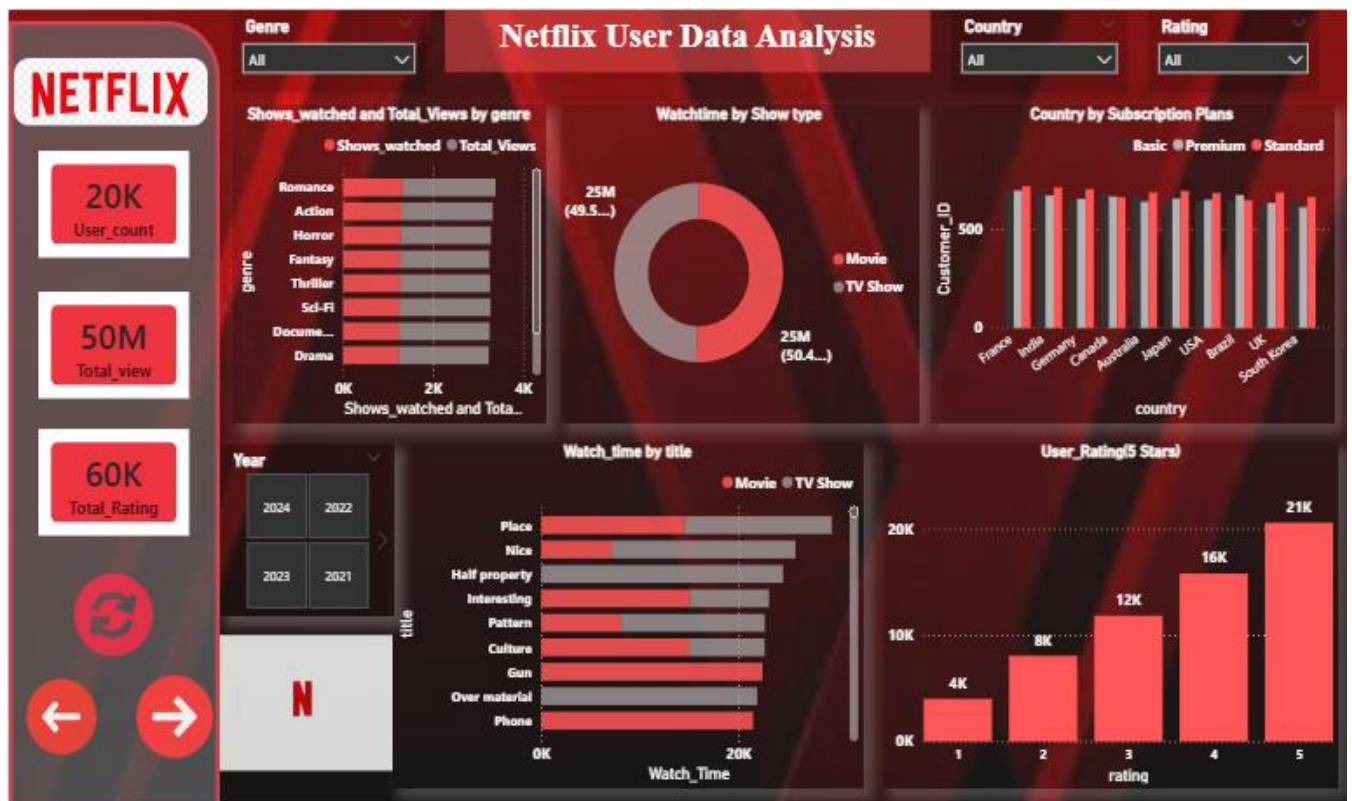
- ✓ Used consistent formatting such as:
 - DD-MM-YYYY or YYYY-MM-DD
- ✓ Removed unwanted spaces and invalid date entries.
- **Ensure rating values are properly categorized.**

The rating column contained different audience categories such as TV-MA, PG-13, R, and others. These values were organized properly for better analysis.

 - ✓ Verified all rating categories for consistency.
 - ✓ Removed duplicate or incorrectly formatted values.
- **Age Group Categorization**
 - ✓ Customers were grouped into different age ranges to understand the distribution of Netflix viewers across various age categories.
- **Analyze the relationship between subscription type**

The Netflix dataset was analyzed to understand how different subscription plans affect total watch time and user engagement.

 - ✓ Basic
 - ✓ Standard
 - ✓ Premium
 - Calculated total watch hours for each subscription plan.
 - Compared average watch time per user.



2.Descriptive & Comparative Analysis

➤ Top 10 Most Watched Shows/Movies

- ✓ Identified the top 10 most-watched Netflix shows or movies based on total watch time.

➤ Genre Popularity by Country

- ✓ Grouped the dataset by
 - Country
 - Genre
- ✓ Calculated the total number of viewers for each genre in every country.
- ✓ Compared viewer counts across genres.
- ✓ Identified the highest-viewed genre for each country.

➤ Distribution of User Ratings (1–5 Stars)

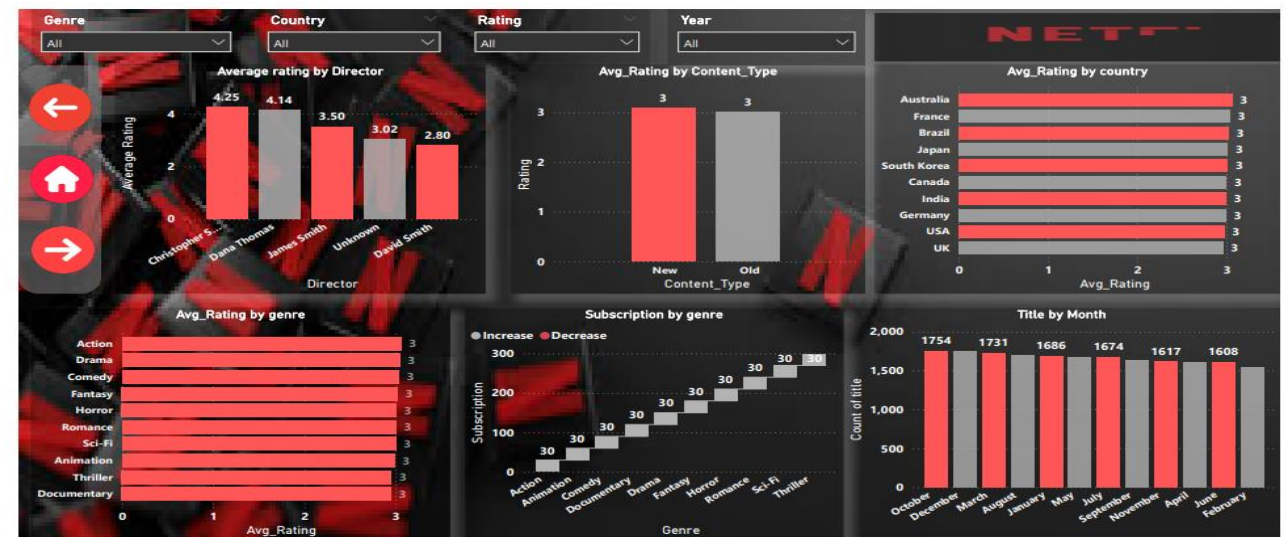
The ratings provided by users were analyzed to understand audience satisfaction levels for Netflix content.(1,2,3,4,5 Stars)

➤ Average rating per genre and per country

- ✓ **Anime** and **Documentary** genres had the highest average ratings.
- ✓ Comedy and Horror genres showed comparatively lower ratings.
- ✓ Drama and Thriller content maintained consistent audience satisfaction

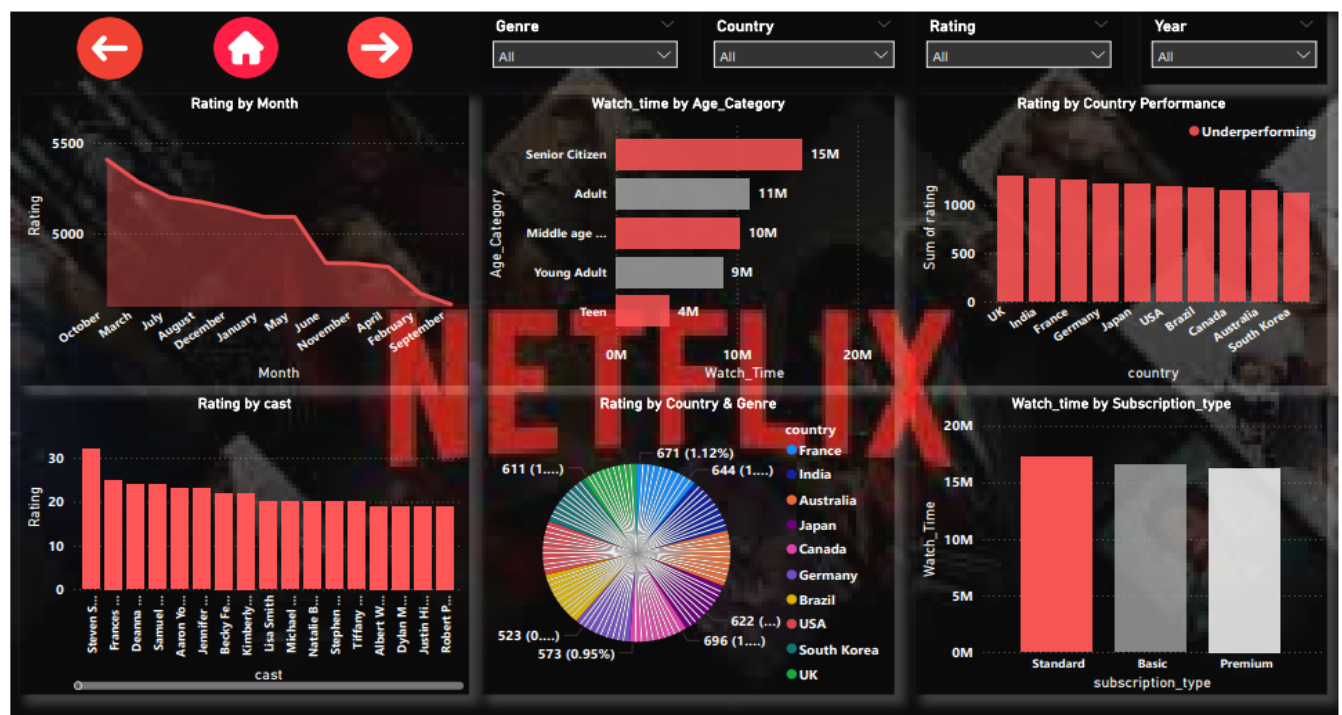
➤ Identifying Recently Added Content

- ✓ Converted the date_added column into proper date format.
- ✓ Applied filters for recent years.
- ✓ Grouped content into:
 - Newly Added Content
 - Older Content



3. Market Insights & Revenue Analysis

- **Subscription Market Share Analysis:**
 - ✓ Basic, Standard, and Premium — based on the total number of subscribers.
- **Find out how many customers watch multiple shows vs. just one show**
 - ✓ Grouped viewing data based on:
 - Customer ID
 - Number of titles watched
 - ✓ Counted the total number of shows/movies watched by each customer.
 - ✓ Categorized customers into:
 - Single-Content Viewers
 - Multiple-Content Viewers
- **Identify which genres get the highest and lowest ratings.**
 - ✓ Grouped Netflix content by genre.
 - ✓ Calculated the average user rating for each genre.
 - ✓ Compared ratings across all categories.
 - ✓ Identified the highest-rated and lowest-rated genres.
- **Compare the ratings of TV Shows vs. Movies.**
 - ✓ Calculated the average user rating for each content type.
 - ✓ Compared audience feedback and engagement levels.



4. Business Decision-Making Scenarios

➤ Director & Cast Influence on Ratings

- ✓ Grouped Netflix content by:
 - Director
 - Cast
- ✓ Calculated:
 - Average user ratings
 - Total watch time
 - Viewer engagement
- ✓ Compared ratings between popular and less-known directors/casts.
- ✓ Ranked directors based on average ratings.

➤ Content Localization Strategy:

- ✓ Countries were evaluated using the following performance indicators (Average Rating, Watch Time and etc)
- ✓ Invest in region-specific movies, TV shows, and documentaries that match local culture, language, and audience preferences.

➤ Monthly Content Addition Analysis

- ✓ Converted the `date_added` column into a proper date format.
- ✓ Extracted:
 - Month
 - Year
- ✓ Counted the number of titles added each month.
- ✓ Compared monthly content additions using charts and dashboards.



Results:

- ✓ Clean and structured datasets ready for analysis.
- ✓ A Power BI dashboard with key insights on content performance and customer behavior.
- ✓ Actionable business recommendations with PPT presentation for Netflix's content and subscription strategy

Technical Tags:

Power BI, Excel, Data Cleaning, Business Intelligence, Customer Analysis, Content Analytics, Data Visualization, EDA.

Skills take away From This Project

- ✓ Data Cleaning and Preprocessing
- ✓ Exploratory Data Analysis (EDA)
- ✓ Data Visualization using Power BI
- ✓ Business Insights and Decision Making
- ✓ Customer Behavior Analysis
- ✓ Content Performance Analytics
- ✓ Subscription and Engagement Analysis
- ✓ Dashboard Creation in Power BI

Conclusion:

The Netflix Data Analysis project provided valuable insights into viewer behavior, subscription trends, content ratings, and regional engagement using Microsoft Power BI. The analysis showed that TV Shows and Premium subscriptions generate higher engagement and watch time. Popular genres like Anime and Drama received better audience ratings, while localized content was important for improving regional performance. Interactive dashboards helped transform raw data into meaningful business insights and supported better decision-making strategies